

Supporting Information for

Temperature impacts on fish physiology and resource abundance lead to faster growth but smaller fish sizes and yields under warming

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Model parameterization

Species-specific parameters

Below follows a description for how we acquired default parameters which are valid at the reference temperature, T_{ref} , where the temperature scaling equals 1 for all rates (see main text for which rates are assumed temperature dependent and how). In ‘mizer’, multi-species size spectrum models (MSSMs), species-specific parameters are required. Parameters that have no defaults and must be provided are: asymptotic size (W), size at maturation (w_{mat}), preferred predator-prey mass ratio (PPMR) (β), the standard deviation of PPMR (σ) and the maximum recruitment in the Beverton-Holt stock recruitment function (R_{max}) (Scott *et al.* 2019).

For cod, we initially acquired the parameters β and σ by calculating the mean and the standard deviation for the distribution of mean individual-level log PPMR (β is then exponentiated whereas σ is defined as the standard deviation of the distribution of log PPMRs), using stomach data (downloaded on 2019.03.20) hosted by the International Council for the Exploration of the Sea (ICES), in the FISH STOMACH database (ICES 2010). These data can be downloaded at <http://ecosystemdata.ices.dk/stomachdata/download.aspx>. We used only samples with fully intact prey (no digestion) among the observations from the area we consider (ICES subdivisions SD 25-29, 32, see Fig. S2) and from the time window we use for calibrating the model (1992-2002). When prey weight was not available, we used the relationship $weight = 0.01 \times length^3$, where length is in cm, to approximate weight in g. However, since stomach content data is not a measure of preference per se, as it is only shows what has been eaten (which also depends on prey availability), these parameters should also be evaluated after model fitting. For sprat and herring, we assumed β and σ to be 1000 and 1.3, respectively, which reflects that forage fish typically feed on prey that are small in relation to their body size throughout ontogeny (Aydin *et al.* 2002; Reum *et al.* 2019).

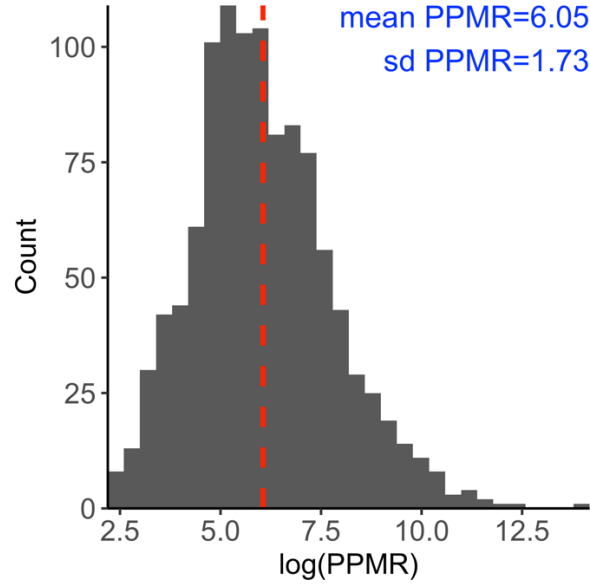


Figure S1. Distribution of individual level predator-prey mass ratio ($\log \text{PPMR}$) for cod. Mean $\exp(\text{PPMR})$ corresponds to β and sd to σ . The dashed red vertical line corresponds to the mean $\log \text{PPMR}$.

From von Bertalanffy (VBGE) growth parameters and theory linking growth to feeding parameters (Hartvig *et al.* 2011), the remaining species-specific parameters can be estimated. The allometric constant in the maximum food intake rate (h_i) is defined as $h_i = \frac{3k_{vb}}{\alpha f_0} W^{1/3}$, where k_{vb} is the Brody growth coefficient of the VBGE, α is the assimilation efficiency, f_0 is the initial feeding level of small individuals and W is asymptotic mass (see main text) (Andersen *et al.* 2009; Scott *et al.* 2019) (but see calibration protocol). When h_i is known, the allometric constant in the search rate function, γ_i , can be calculated (see equation 9 in main text). The allometric constant of standard metabolism, $k_{met,i}$ is by default $0.12h_i$ (but see calibration protocol).

We estimated VBGE parameters k_{vb} and W for cod, herring and sprat in the Baltic Sea using data from the Baltic International Trawl Survey (BITS), maintained by ICES. They are publicly available at the DATRAS database (<http://www.ices.dk/marine-data/data-portals/Pages/DATRAS.aspx>). We downloaded the full data for the years 1992-2002 (the calibration time period) on 2018.11.24, and applied further processing in R using the ‘tidyverse’ packages (Wickham *et al.* 2019). We followed the approach in Ogle (2013) and fit the VBGE using the form $L_a = L_\infty(1 - e^{-k_{vb}(a-t_0)})$ when estimating parameters. Here L_a is the expected length at age a , L_∞ is the average asymptotic length, k_{vb} is the Brody growth rate coefficient and t_0 is a modeling artifact that represents the time or age when the average length

would have been zero. Parameters were estimated using non-linear least squares regression ('nls' package in R), and the packages 'FSA' (Ogle 2018) and 'FSAdata' (Ogle 2017). We fit VBGE parameters using length-at-age data and convert between length and weight using the equation $w = aL^b$, where parameters a and b are estimated in this study from a subset of the BITS data where both weight and length information is available for the same individual. Length-at-maturation was taken from the literature and was converted to weight-at-maturation using length-weight relationships estimated in this study from BITS (Table S1). All models were fitted using R (R Core Team 2020), and for all statistical models we verified through visual inspection that assumptions about error homoscedasticity and normality were met.

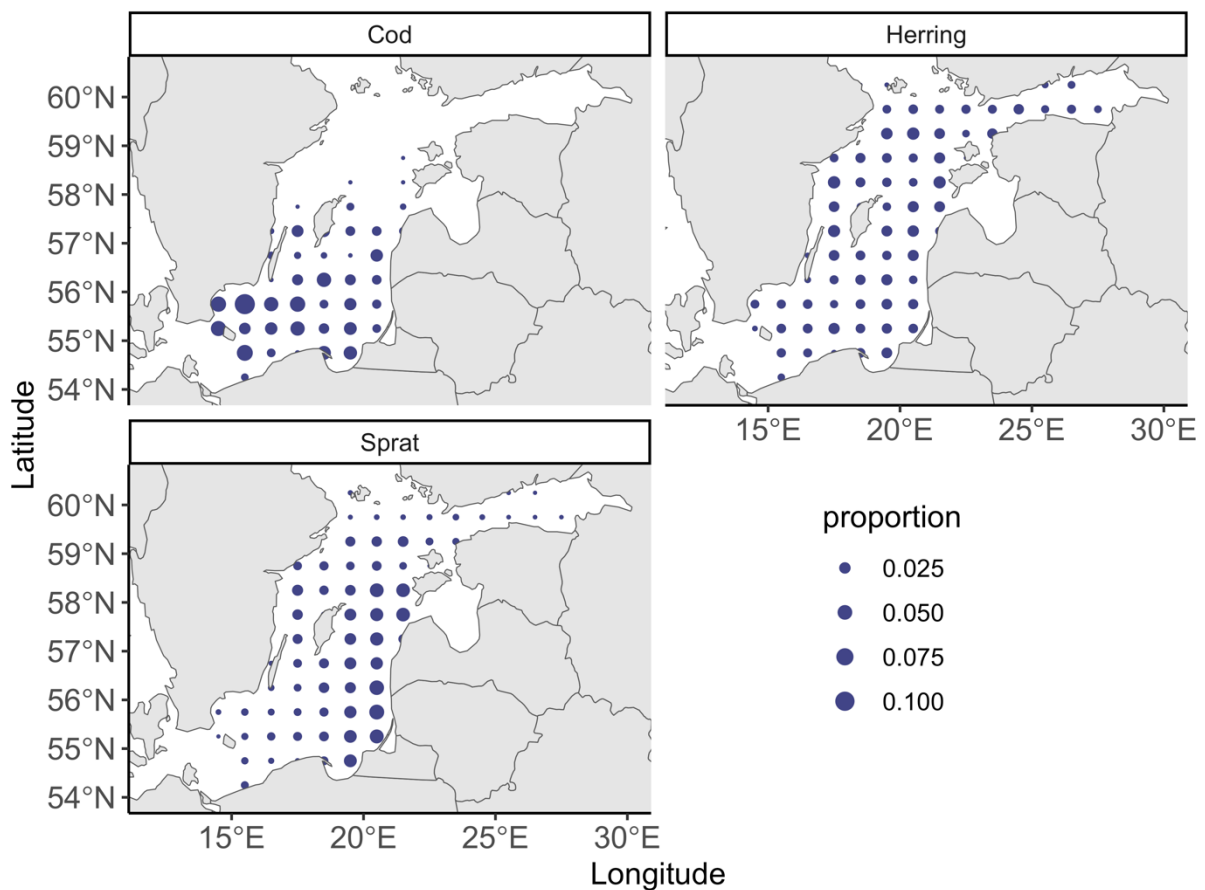


Figure S2. Map of the study area (ICES sub-divisions 25-29+32). Point size is proportional to abundance per ICES rectangle during the calibration time period (1992-2002), using data from the Baltic International Trawl Survey (BITS) for cod, and the Baltic International Acoustic Survey (BIAS) for sprat and herring.

Table S1. Species-specific parameters used in model. Source: 1 = estimated in this study, 2 = (Reum et al. 2019), 3 = (Hartvig et al. 2011), 4 = generalized values based on (Casini et al. 2004), 5 = calculated internally. Parameters in bold are tuned in the calibration process, parenthesis show default values pre-calibration.

Symbol	Description	Unit	Cod	Herring	Sprat	Source
w_{mat}	Maturation size*	g	267	17	4	1
W	Asymptotic weight	g	44903	125	21	1
k_{vb}	Brody growth coefficient	yr ⁻¹	0.07	0.165	0.287	1
t_0	Age at length 0	-	-0.94	-3.74	-2.97	1
β	Preferred predator prey mass ratio	-	426	1000	1000	1,2
σ	Width of prey size preference	-	1.7	1.3	1.3	3
h	Constant for max. food intake	g ¹⁻ⁿ yr ⁻¹	27 (20.7)	8.9 (6.9)	8.6 (6.6)	5
γ	Constant for volumetric search rate	g ^{-q} m ² yr ⁻¹	0.6 (0.38)	0.18 (0.11)	0.17 (0.1)	5
k_{met}	Constant for metabolic rate (note this term is assumed to represent all metabolic costs, e.g., standard and for activity)	g ^{1-p} yr ⁻¹	2.49	0.83	0.22	5
$\epsilon, \epsilon_{repro}$	Reproductive efficiency	-	5e-05 (0.01)	8e-04 (0.01)	1e-03 (0.01)	1
a	Constant in length-weight relationship	-	0.0078	.0042	.0041	1
b	Exponent in length-weight relationship	-	3.07	3.14	3.15	1
R_{max}	Maximum recruitment in the Beverton-Holt stock recruitment function**	g/m ²	0.003 (0.0826)	0.1 (11.1)	1.16 (7.3)	-
$\theta_{i,ben}$	Benthos availability	-	0.5	0.5	0	4
$\theta_{i,pla}$	Plankton availability	-	0.5	0.5	1	4

*Length at maturation for cod and sprat are taken from van Leeuwen et al. (2013) (30 cm and 9 cm, respectively) and for herring we used Huss et al. (2012) (14 cm), based on (Vainikka et al. 2009a, b). **See calibration protocol for unit and volume scaling

Table S2. General parameters. Source: 1 = (Scott et al. 2019), 2 = (Hartvig et al. 2011), 3 = (Blanchard et al. 2014), 4 = this study, 5 = Audzijonyte et al., unpublished.. Parameters in bold are tuned in the calibration process, parenthesis show default values pre-calibration.

Symbol	Description	Value	Unit	Source
α	Assimilation efficiency	0.6	-	1,2
w_0	Egg weight	0.001	g	1,2
f_0	Initial feeding level	0.6	-	1,2
n	Exponent of max. consumption	2/3	-	1,2
m	Exponent to $\frac{w}{w_{mat,i}}$ in smooth maturation function	5	-	5, 4
q	Exponent of search volume	0.8	-	3
p	Exponent of metabolic rate	0.7	-	3
μ_0	Pre-factor for background mortality	0.6	yr ⁻¹	1
λ	Exponent of background resource spectra	$-2 - q + n$	-	2
r_0	Regeneration rate of background resource spectra	4	g ¹⁻ⁿ yr ⁻¹	4
κ	Carrying capacity of background resource spectra (benthos, plankton)	9 (11)	g^{$\lambda-1$}m²	4
$w_{cut,P}$	Cut-off size of plankton size spectrum	1	g	4
$w_{cut,B}$	Cut-off size of benthic size spectrum	20	g	4

Temperature dependence

Table S3 Parameters of distributions describing activation energies of temperature-dependent rates in the size spectrum model. See Figure S11 for 200 random draws from these normal distributions that were used as input in projections.

Symbol	Rate	Distribution (mean, s.d)	Source
A_{met}	Metabolism	$Normal(0.62, 0.025)$	(Lindmark <i>et al.</i> 2022)
A_{mor}	Background mortality	$Normal(0.62, 0.025)$	(Pauly 1980; Brown <i>et al.</i> 2004; Lindmark <i>et al.</i> 2022)
A_{int}	Maximum consumption; Search volume*	$Normal(0.69, 0.078)$	(Lindmark <i>et al.</i> 2022)
A_{gro}	Background resources regeneration rate	$Normal(0.73, 0.1)$	Experimental data from Savage <i>et al.</i> (2004) for algae, phyto- and zooplankton.
A_{car}	Background resource carrying capacity	$Normal(-0.8, 0.1)$	Assumed to be $-A_{gro}$, based on metabolic arguments and constant resources (Savage <i>et al.</i> 2004; Gilbert <i>et al.</i> 2014; Bernhardt <i>et al.</i> 2018)

* For simplicity we assume maximum consumption to scale the same way with temperature as search volume, and the parameter estimates come from maximum consumption experiments.

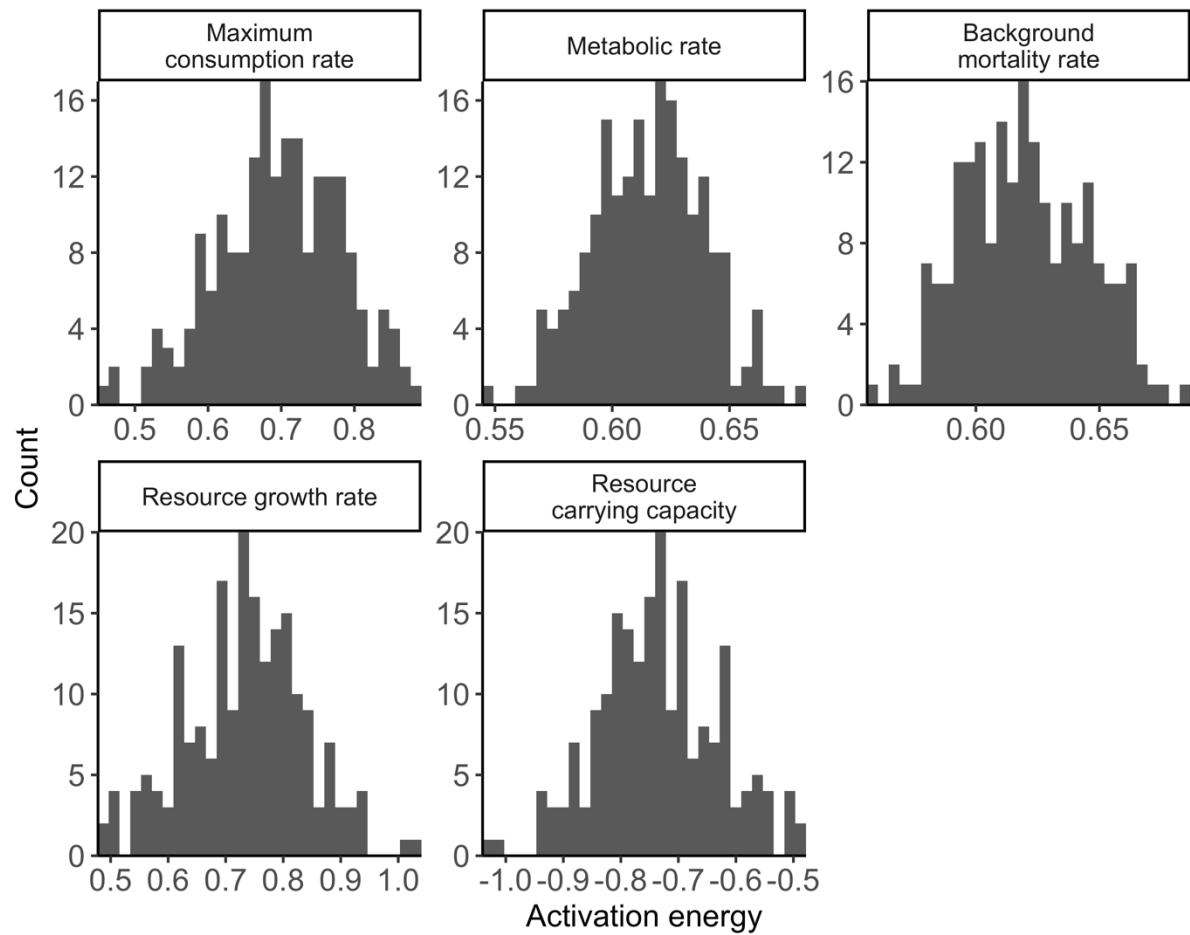


Figure S3. 200 random samples from the distributions describing the activation energies of the rates resource carrying capacity and resource growth rates (assumed same for benthic and pelagic background resources), metabolic rate, background mortality rates and maximum consumption rates. See Table S3. Combinations of these parameters were used in model projections to understand how uncertainty in these parameters affect individual- and population level metrics.

177 Model calibration and validation

178 Calibration protocol

179 We calibrated to the model using mean SSB and F between 1992-2002 from stock assessments,
180 which were acquired from ICES Working Group Reports (ICES 2013, 2015), and are in unit
181 10^6 kg/area. This area corresponds to ICES subdivisions 25-29+32, which equals
182 approximately 2.49×10^{11} m² (estimated using ICES shapefiles) (ICES 2021). We first defined
183 the size spectrum model in units g/m² and then used the conversion factor 249 to express
184 biomasses in unit 1000 tonnes/area. Below follows a step-by-step description of how the model
185 was calibrated to the assembled Baltic Sea data after being initially parameterized.

- 186 1. Determine a starting value for plankton and benthos κ . The aim is to find coexistence
187 of the three fish species and for their spawning stock biomasses (SSBs) to be within an
188 order of magnitude of SSBs from stock assessment. Prioritize coexistence over fits of
189 SSBs, as these are calibrated later. As starting values for $R_{max,i}$, which ensures
190 coexistence, we used the same ratio between $R_{max,i}$ and κ as calibrated in (Blanchard
191 et al. 2014), expressed in unit m² (by dividing with 10^{11} , which corresponds roughly to
192 the area of the North Sea). This procedure resulted in a starting value for κ_B and κ_P of
193 11, in unit g ^{$\lambda-1$} m².
- 194 2. Evaluate body growth rates against empirical data.
 - 195 a. If modelled body growth rates are low, check the species-specific feeding levels
196 $f_i(w)$. The feeding level describes the level of satiation, with 0 being unfed and
197 1 completely satiated. For reference, a feeding level of 0.2 is the minimum to
198 cover basic metabolic costs (by default) and thus does not allow for body
199 growth, whereas a constant feeding level of 0.6 fits a von Bertalanffy curve
200 (Andersen et al. 2009; Scott et al. 2019). If low feeding level seem likely as a
201 cause for poor growth in the model, check the search rate (γ) or available prey.
202 If not, check the net energy acquisition.
 - 203 i. Net energy acquisition is largely determined by maximum consumption
204 and metabolic losses. Earlier explorations suggest modelled growth
205 rates can be lower than observed (Scott et al. 2019) with default k_{met}
206 and h (constants in allometric metabolism and maximum consumption

- rates, respectively), so these may need to be modified based on empirical data or other theoretically derived relationships.
- ii. Go back to step 1. With altered k_{met} and/or h , κ needed for coexistence and fit to SSB may change. Therefore, a κ can be chosen that puts modelled SSBs closer to SSBs from stock assessment, while still allowing for coexistence and maintaining modelled growth rates in line with observed.
3. If κ allows for coexistence and results in SSBs within an order of magnitude to SSBs obtained from stock assessments and body growth rates are realistic, tune R_{max} to match SSBs from the size spectrum model closer to that from the stock assessment. This is done by finding the vector of R_{max} -values that minimizes the residual sum of squares between the two SSBs.
 4. Verify that SSBs and growth rates are still close to empirical data, after optimizing R_{max} . Then evaluate the ratio of egg production before and after density dependence is added to the Beverton-Holt type stock-recruit curve (RDI/RDD).
 - a. If RDI/RDD is high (≈ 100) (Jacobsen *et al.* 2017), and RDD is close to R_{max} , consider lowering $erepro$, because the high RDI/RDD indicates that species are resistant to fishing (flat yield~fishing mortality curves).
 - i. Go back to Step 2 and proceed forward from there.
 5. Verify that SSBs, growth rates and RDI/RDD are realistic. Then assess emergent diets.
 - a. If predatory interactions do not match independent stomach data, re-evaluate the predation kernel. We judged this for cod based on (Niiranen *et al.* 2019), who show that benthos (*Saduria entomon*) are found in stomachs of small cod whereas sprat and herring start to appear in the size classes of 20-29 cm. We assume sprat and herring feed mostly on background spectra with only limited piscivory and cannibalism.
 6. Project forward and backwards with time-varying fishing mortality from stock assessment output and temperature data. Compare qualitative trends with stock assessment data. Since the stock assessment data for the three fish species are outputs from different models and assumptions, this serves more as a useful comparison rather than validation of the model.

Results from calibration procedure

The following figures show model output from the final set of parameters found through the calibration procedure described above (Table S1-S2). In addition to the bold parameters in Tables S1-S2 (ϵ and R_{max}), we also updated the constant in the maximum consumption rate (h) with a factor of 1.3 as body growth was otherwise poor despite a normal feeding level (~ 0.6). The level of density dependence imposed by the stock-recruit function (see Eq. 14-15) was evaluated by assessing the ratio of the physiological recruitment, $R_{phy,i}$, to the recruitment R_i (RDI/RDD in the *mizer* package) (Jacobsen *et al.* 2017). If this ratio is 1, there is no additional density dependence from the stock-recruitment function because the species are at the linear part of the stock-recruit relationship. After calibrating the model, we acquired $\frac{R_{phy,i}}{R_i}$ ratios of 554, 79 and 28 for cod, herring and sprat, respectively. High values (>100) suggest recruitment is largely controlled by the external parameter R_{max} (Jacobsen *et al.* 2017), which in our case can be said for cod, and to a lesser extent, sprat and herring.

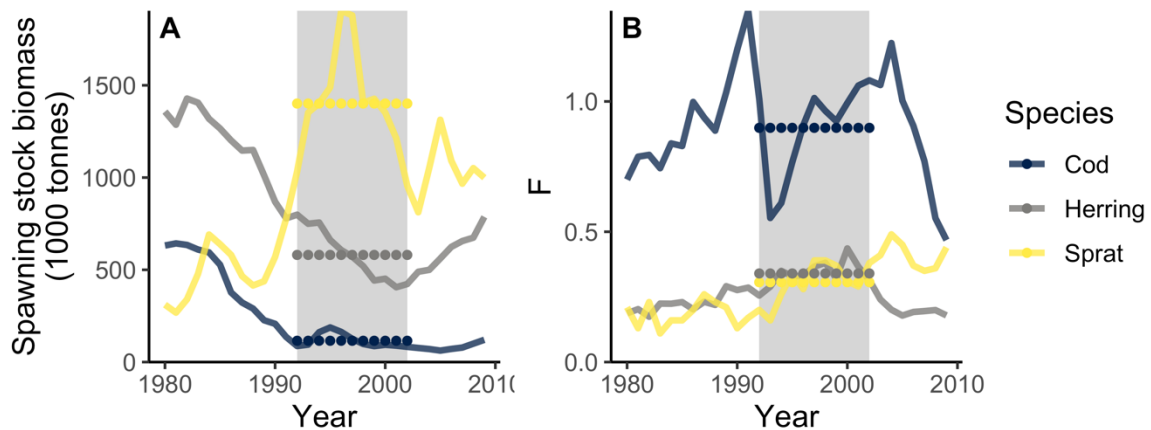


Figure S4. Time-series of spawning stock biomass (SSB) (A) and fishing mortality (B) from stock assessment model estimates (ICES 2013, 2015). Grey background shows calibration period, dotted horizontal lines show the mean of each time series in the calibration time period.

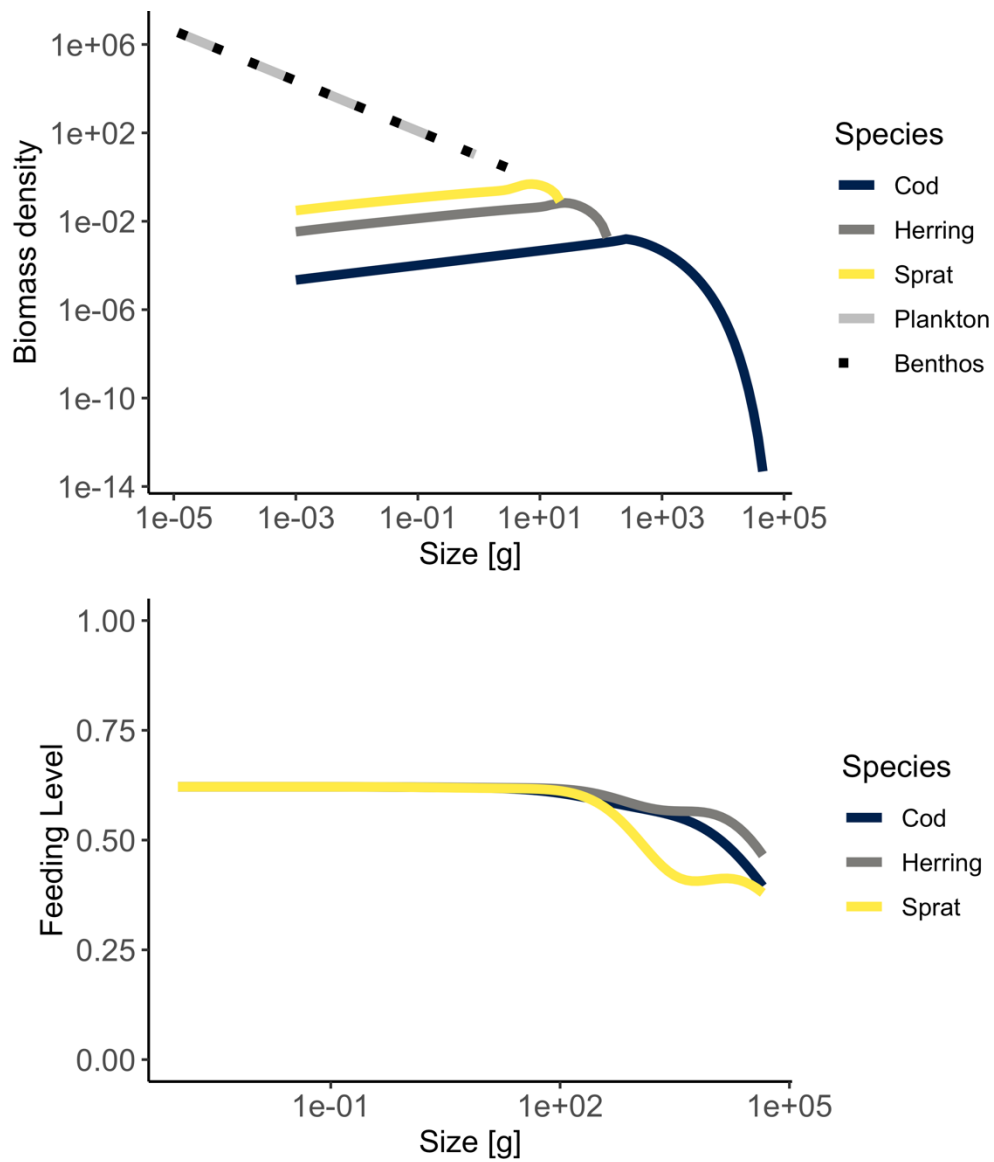


Figure S5. Biomass spectra of species and background resources (top) and feeding levels as a function of body size (g) from the size spectrum model simulated to steady state using the calibrated model parameters (Table S1-S2).

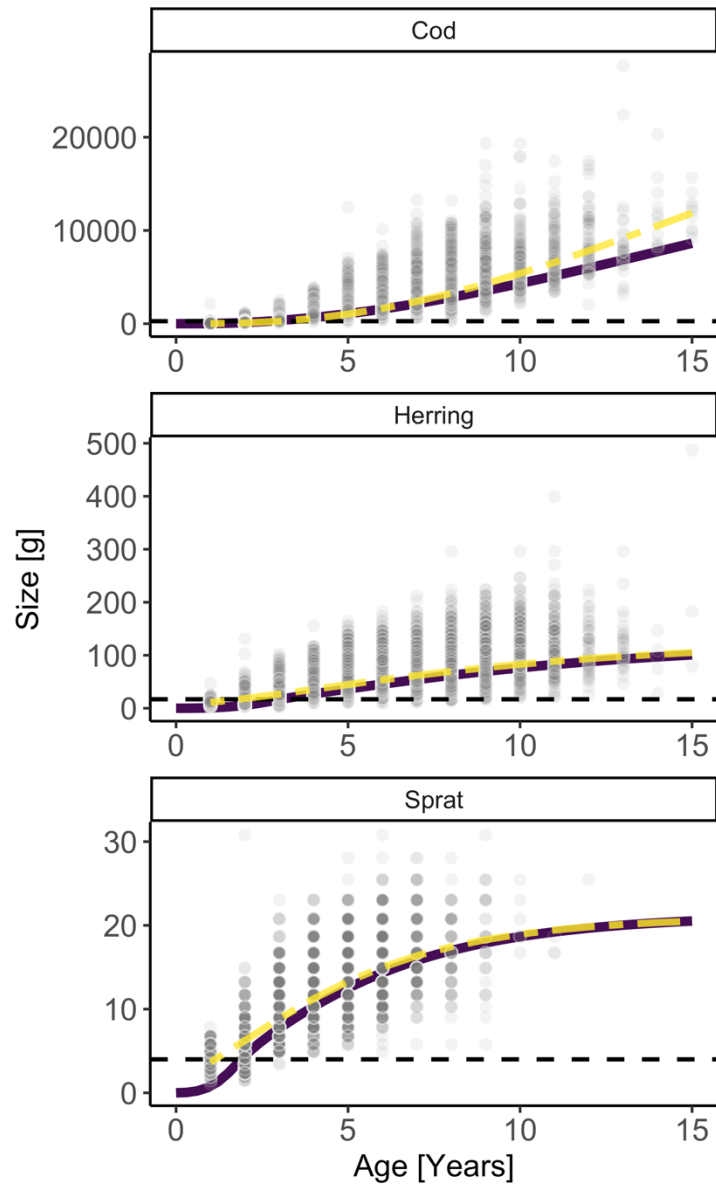


Figure S6. Size-at-age (g) from size spectrum model (solid purple line) run to steady state using the calibrated model parameters, growth curves from the von Bertalanffy growth equation (two-dashed yellow line) fitted to length-at-age data from the Baltic International Trawl Survey (semitransparent dark grey points) and then converted to mass using the length-weight relationship estimated in this study.

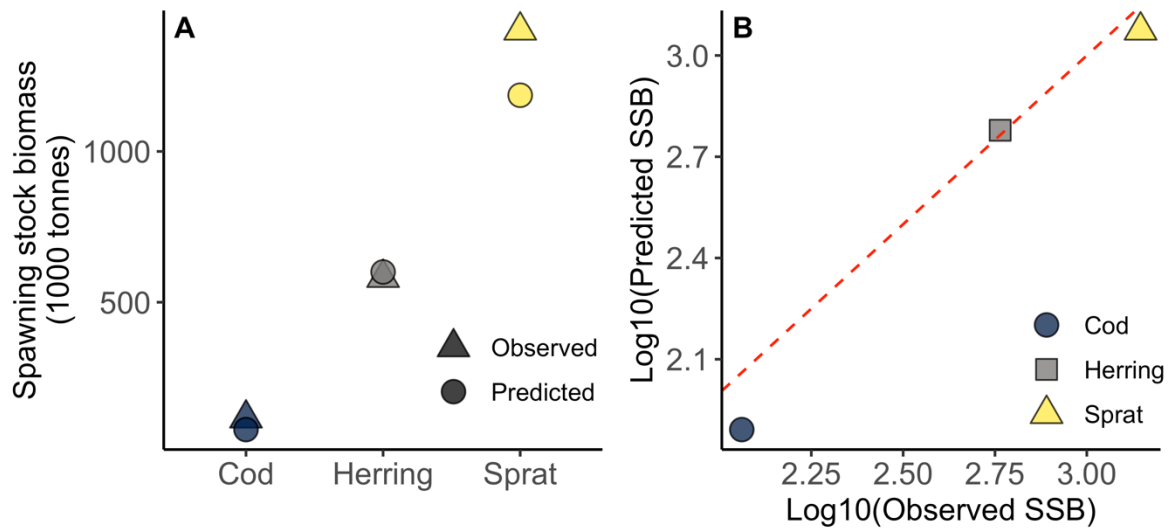


Figure S7. Predicted (size-spectrum model) vs 'observed' (stock assessment) spawning stock biomass (SSB) per species. A) Mean predicted SSB (final 20 time steps of projection) for each species and observed SSB in the calibration period. B) Same data on log10 scale plotted against each other with a 1:1 line (red dashed).

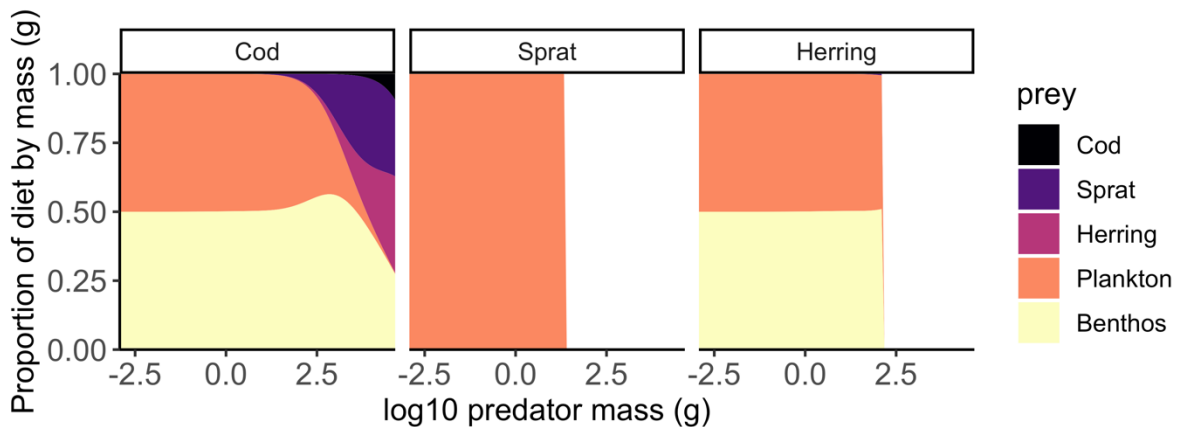


Figure S8. Proportion of diet by mass as a function of predator body mass at steady state. Cod becomes piscivorous when approximately 25cm (Niiranen et al. 2019). This roughly corresponds to a mass of about 150g, which is slightly smaller than what emerges from the model (~300 g).

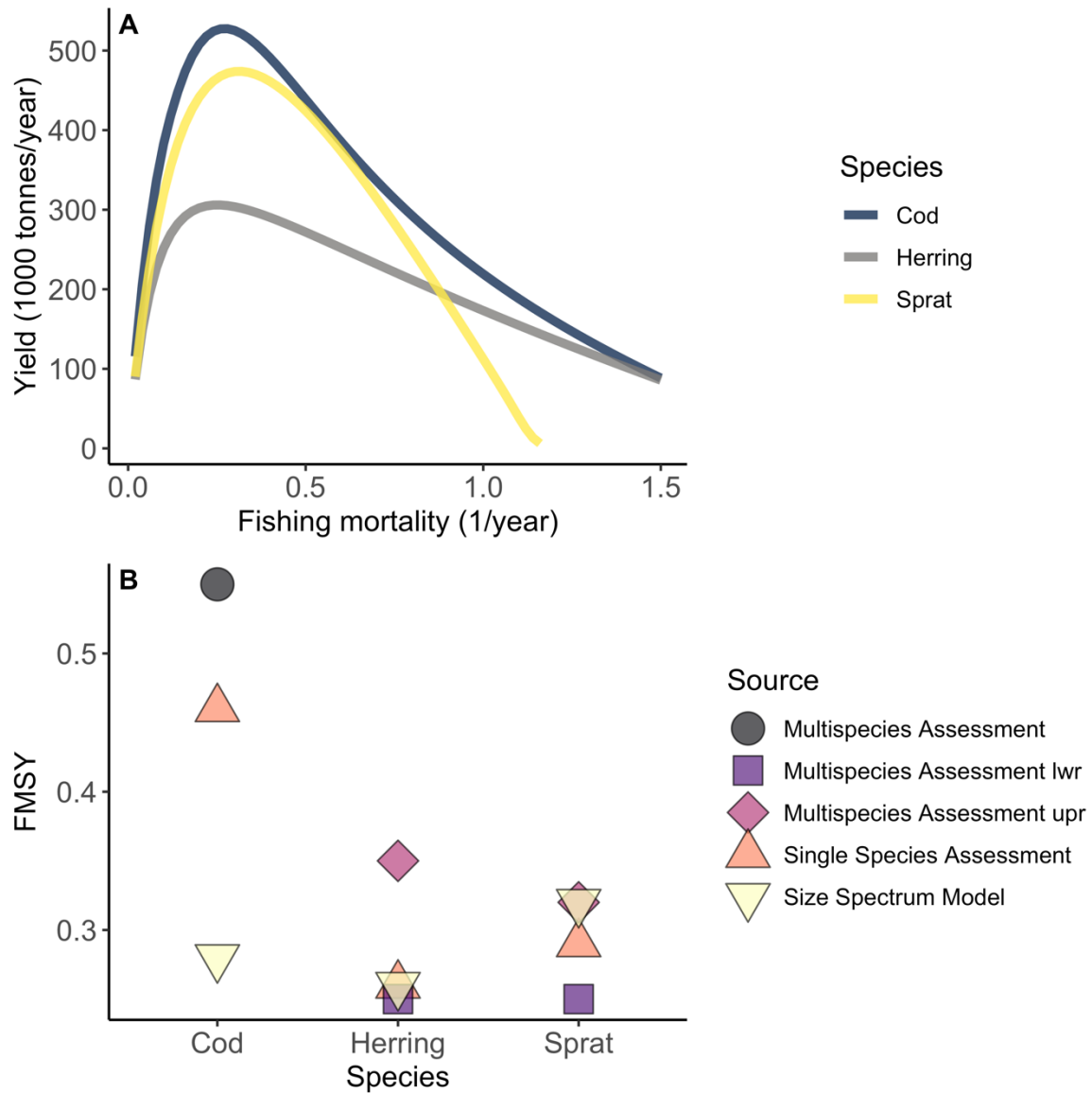


Figure S9. (A) Yield in the size spectrum model (estimated by keeping each species at their mean assessment F_{MSY}) at the reference temperature (T_{ref}) and (B) corresponding F_{MSY} (fishing mortality resulting in the highest long-term yield) for each species in the size spectrum model compared to those obtained in multi- and single-species stock assessment models (ICES 2013, 2015), using default parameters.

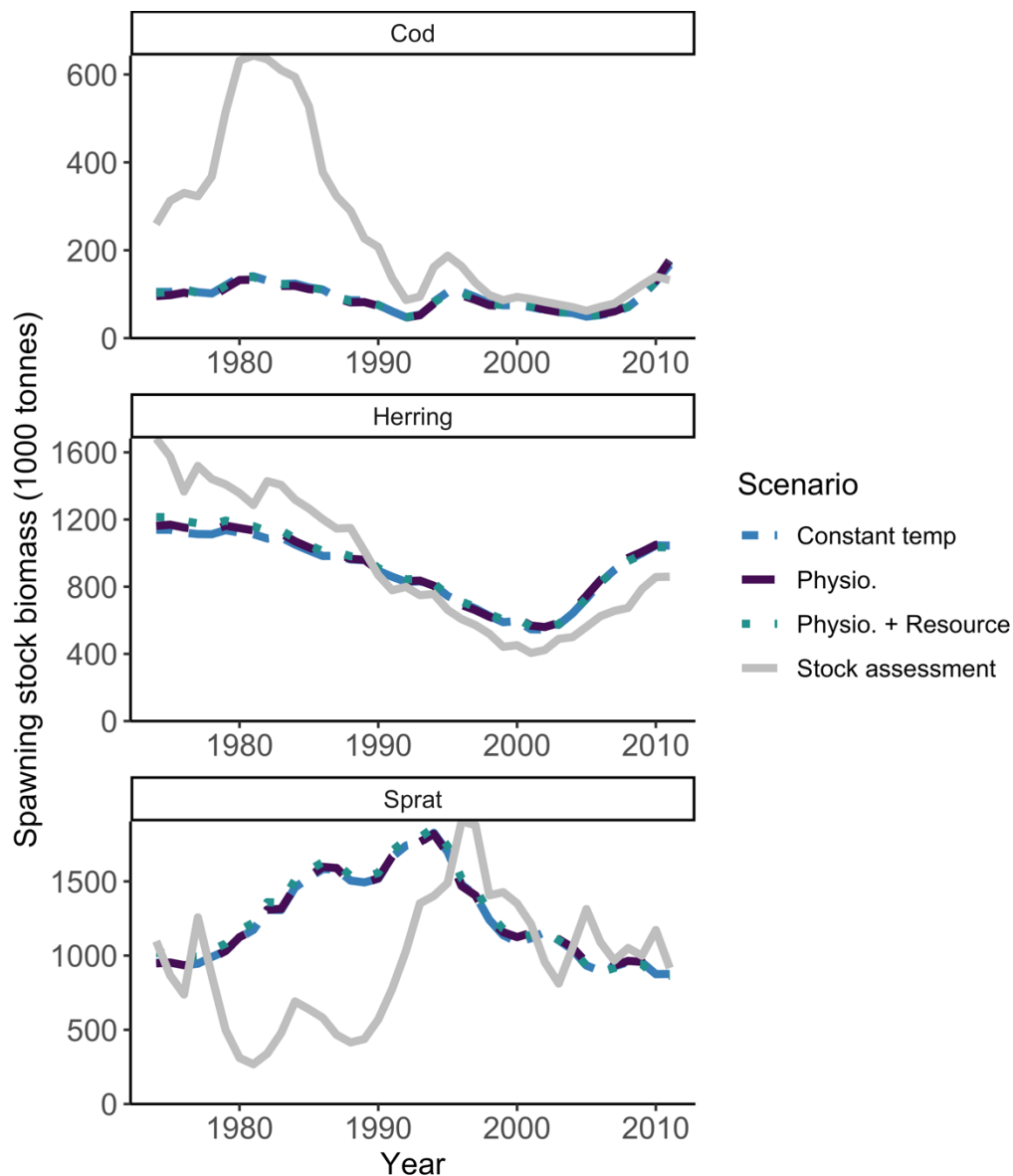


Figure S10. Temporal development of spawning stock biomass (SSB). Temperature varies in the time series according to the RCA4-NEMO model using the RCP 8.5 scenario. The projections start in 1874 (not shown) to allow a 100-year burn-in period. This is sufficient to reach steady state before historical fishing efforts are introduced (in 1974-2012). F-values by species in the burn-in period are equal to the values in 1974 (first year of effort-time series). The solid grey lines are SSB from stock assessments, the two-dashed blue lines are SSBs from the size-spectrum model projections assuming constant temperatures (equivalent to no temperature effects). The purple dashed line is the model where only physiological processes are temperature dependent, the dotted teal line is where physiological processes and resources are temperature dependent. Mean activation energies are used (Table S3).

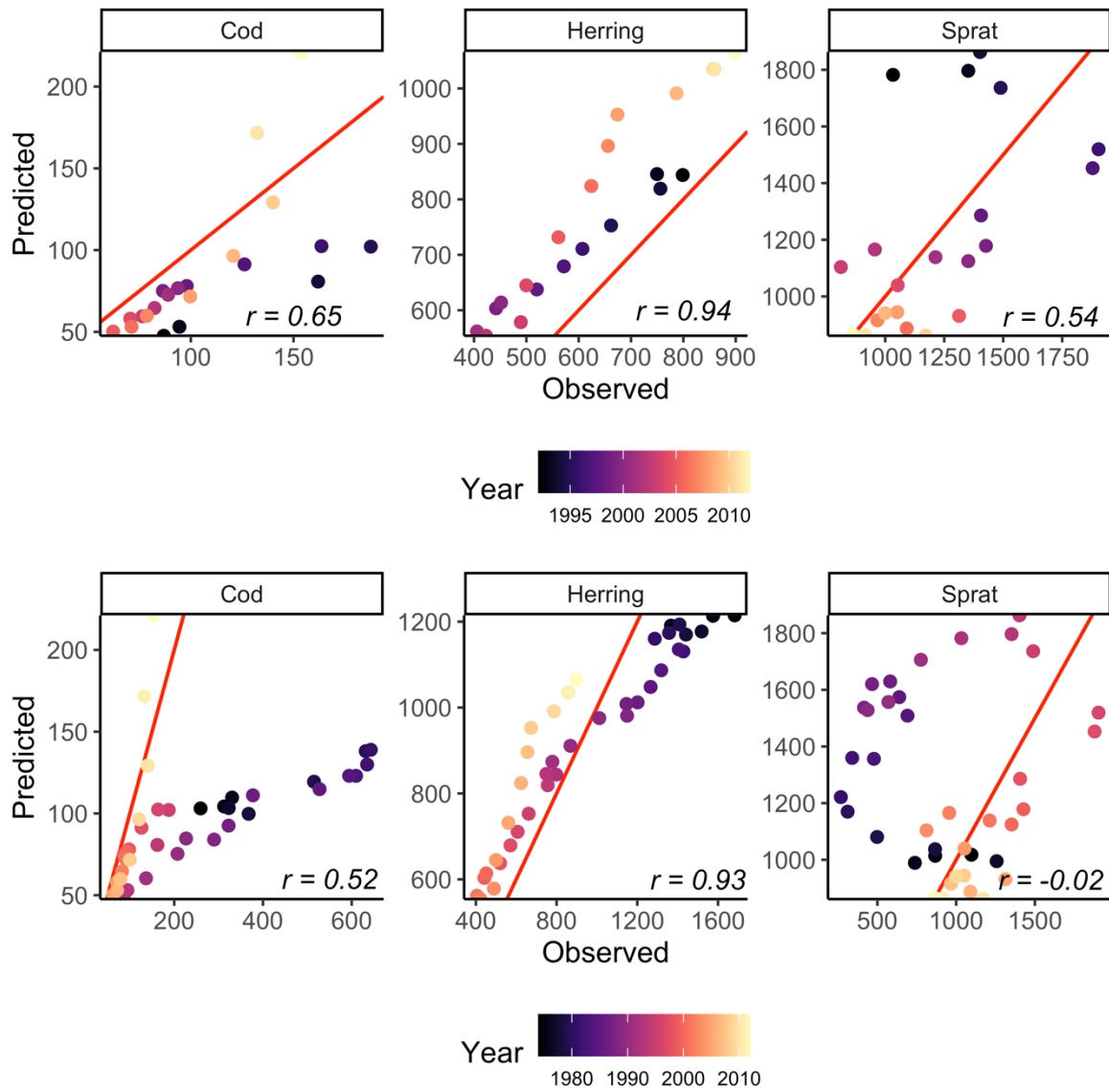
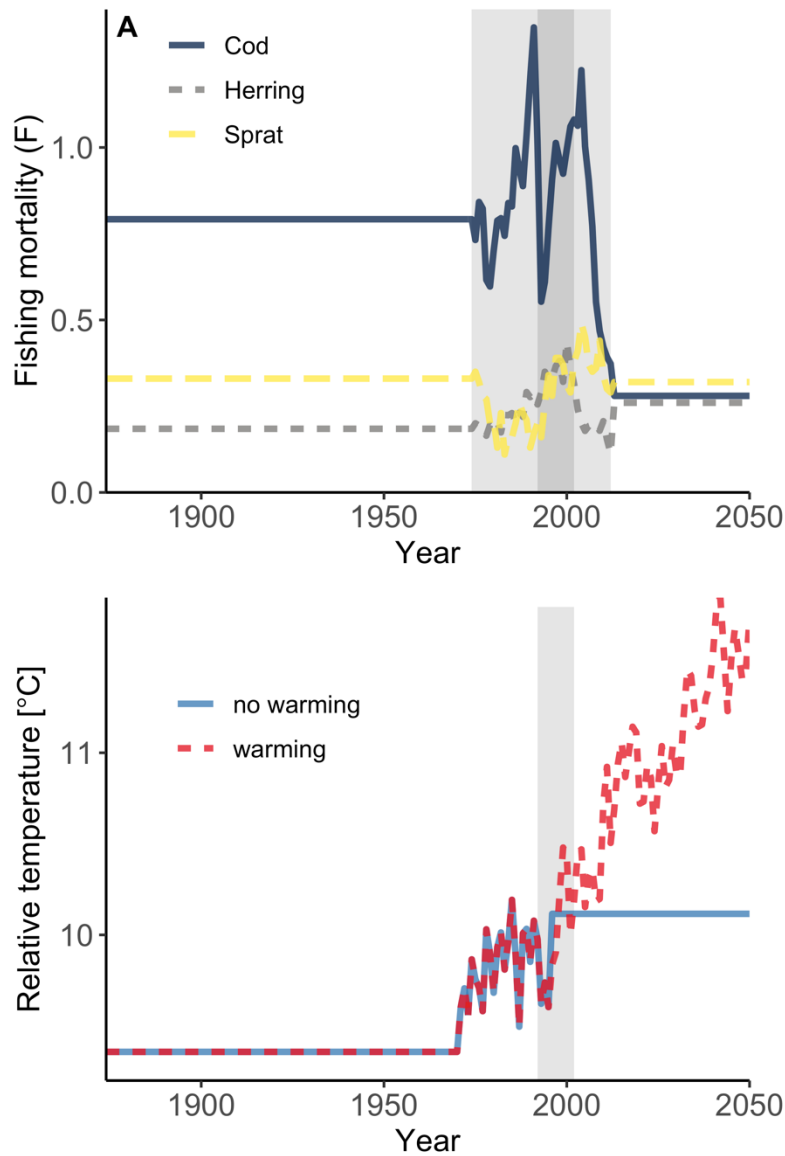


Figure S11. Correlation between SSB from size-spectrum model (“predicted”) and stock assessment models (“observed”) between 1992-2012, i.e., from the start of the calibration to the last year (top row) and 1974-2012, i.e., from the start of the time series (bottom row). The Pearson correlation coefficients are indicated in the bottom right corner. Colors depict year.

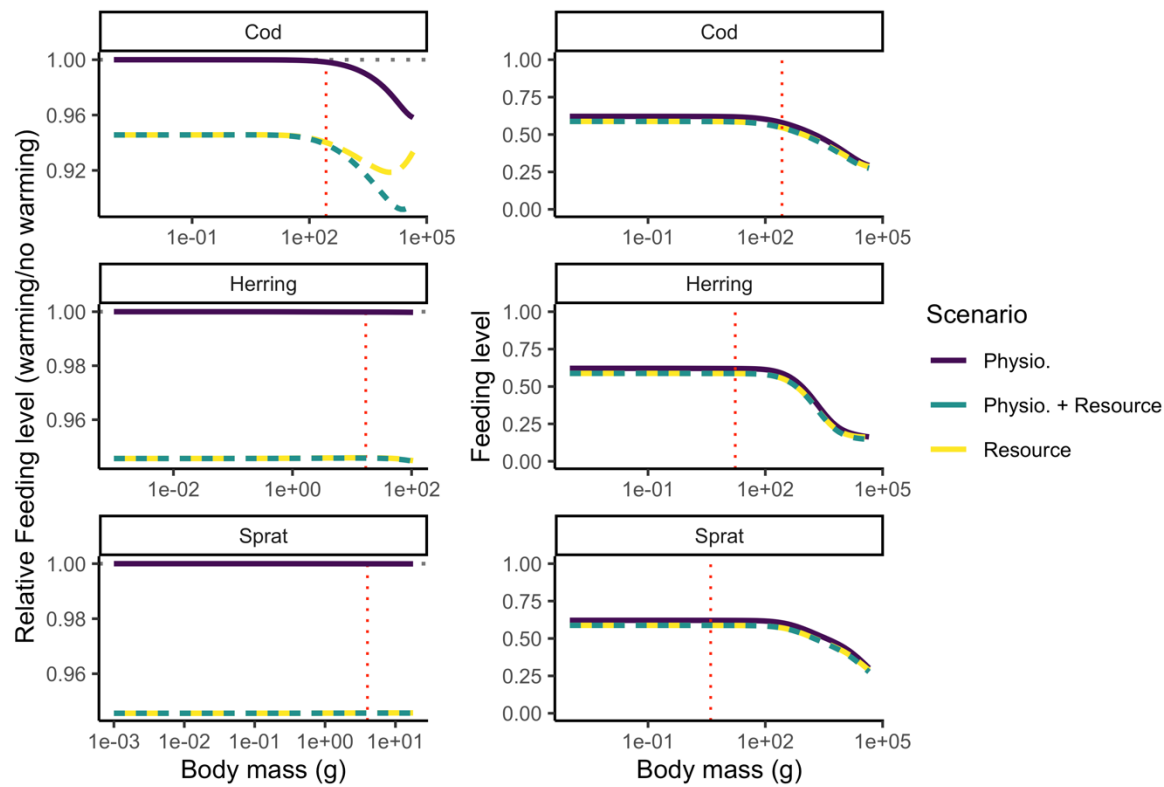


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338 *Figure S12. Time series of (A) fishing mortality by species and (B) relative temperature. For*339 *fishing mortality (A) the light grey area corresponds to years where mortality values are*340 *estimated from stock assessments. The dark grey area is the model calibration time period. We*341 *treated the 100 years prior to the start of the F -time series as burn-in and kept the F constant*342 *and equal to the first F time series (in year 1974). The future projections correspond to the*343 *F_{MSY} estimated in the size spectrum model (estimated by keeping each species at their mean*344 *assessment F_{MSY}). The relative temperature is scaled by adding a constant of 10.11562 to the*345 *relative change in sea surface temperature from the regional coupled model system RCA4-*346 *NEMO using the RCP 8.5. This constant makes the mean temperature in the calibration period*347 *the same as the reference temperature, T_{ref} (i.e., no temperature effects). Years prior to the*348 *temperature model are assigned the first value in the temperature record. In all time-varying*

temperature projections, the initial temperature is the same. The no-warming scenario gets a constant temperature equal to the mean temperature in the calibration time-period in 1997 (blue solid line, mid-year in calibration time window) while the warming scenario continuous along the temperature projection from the regional coupled model system RCA4-NEMO using the RCP 8.5 scenario (red dashed line).

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385 *Figure S13. Feeding levels from model projections to 2050 assuming warming according to*
386 *RCP 8.5 while keeping fishing mortality at average F_{MSY} levels (as estimated at T_{ref}). Left*
387 *column shows feeding levels relative to a non-warming simulation (i.e., dashed horizontal grey*
388 *line) and right column shows absolute feeding level. Activation energies are the means of their*
389 *respective distributions (no uncertainty).*

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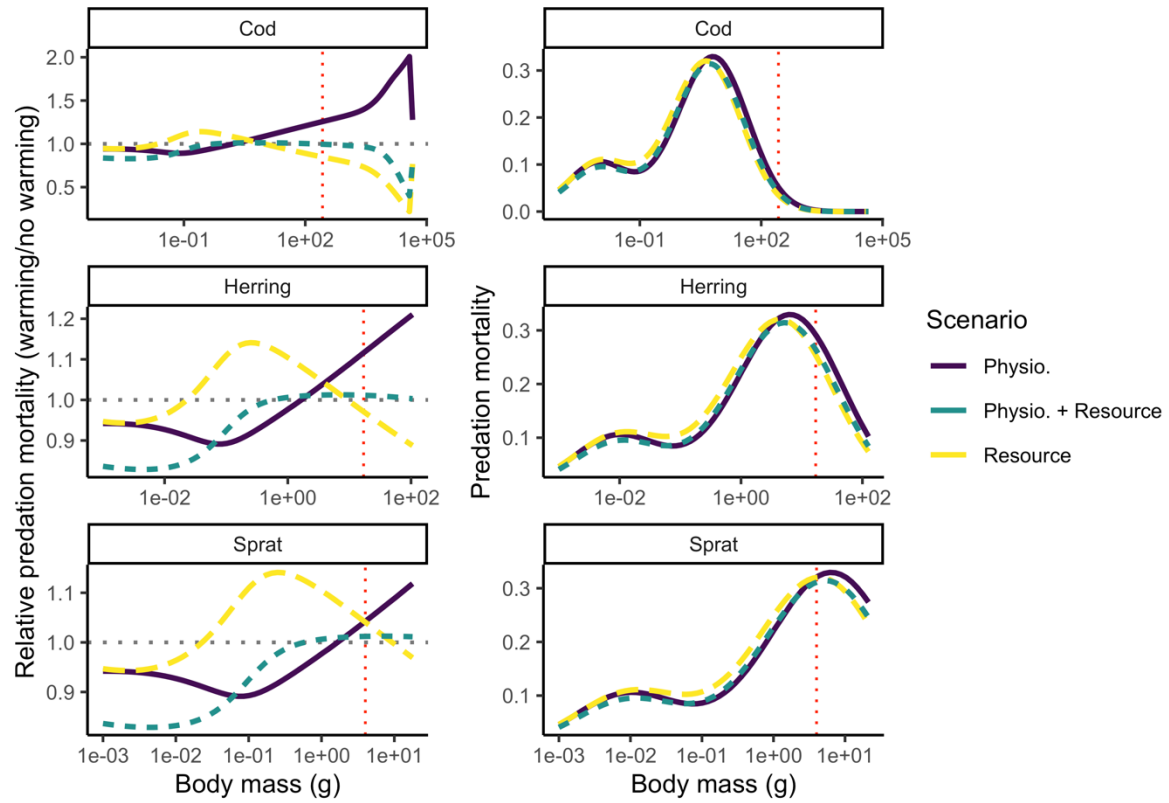
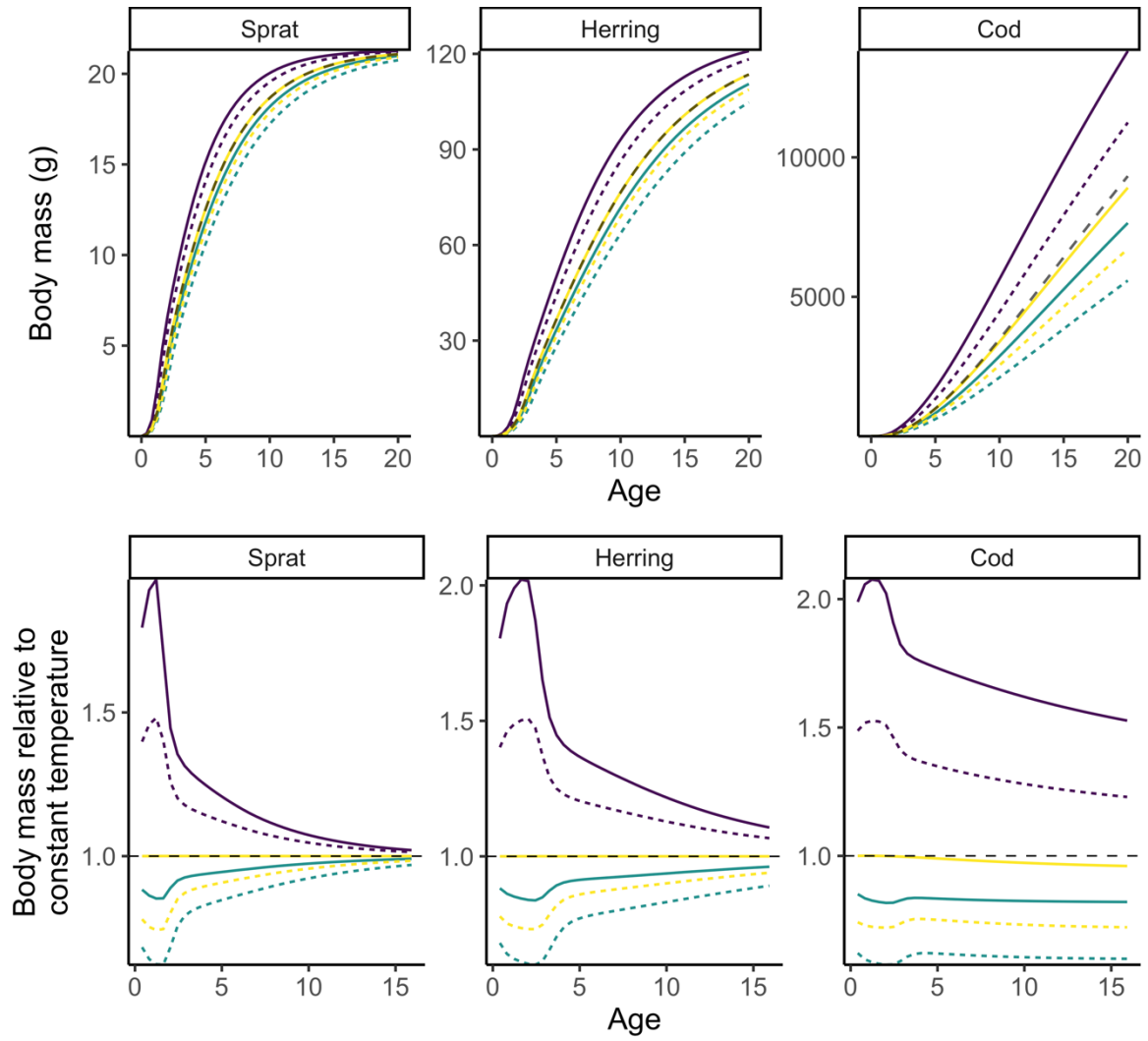


Figure S14. Feeding levels from model projections to 2050 assuming warming according to RCP 8.5 while keeping fishing mortality at average F_{MSY} levels (as estimated at T_{ref}). Left column shows predation mortality levels relative to a non-warming simulation (i.e., dashed horizontal grey line) and right column shows absolute predation mortality. Activation energies are the means of their respective distributions (no uncertainty).



— int — met — mor — no resource temp dep - - - resource temp dep

Figure S15. The effect of including temperature dependence for one physiological process for growth with temperatures $T_{ref} + 2^{\circ}\text{C}$. Here we include temperature-dependence for one physiological process at the time (purple=only intake rates, green=only metabolic rates, yellow=only mortality rates), with and without temperature dependence on resources (dashed and solid lines, respectively). Top panel depicts size-at-age and the bottom panel depicts size-at-age relative to size-at-age from a projection at T_{ref} non-temperature dependent model.

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